# FIITJEE

# ALL INDIA TEST SERIES

## CONCEPT RECAPITULATION TEST – I

## JEE (Advanced)-2019

**PAPER – 1**

# ANSWERS, HINTS & SOLUTIONS

## *Physics*

**PART – I**

**SECTION – A**

1. AC

Sol. $-\frac{1}{F} = P = 2P_{l_1} + 2P_{l_2} + P_m$ …(1)

$P_{l_1} = \frac{1}{f_1} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$

$P_{l_1} = [(1.5 - 1]\left[-\frac{1}{10} - \frac{1}{15}\right] = -\frac{1}{12}$ …(2)

$P_{l_2} = \frac{1}{f_2} = (\mu - 1)\left[\frac{1}{R_1} - \frac{1}{R_2}\right]$

$P_{l_2} = \left(\frac{4}{3} - 1\right)\left[\frac{2}{15}\right] = \frac{2}{45}$ …(3)

$P_m = -\frac{1}{f} = +\frac{2}{15}$ …(4)

$-\frac{1}{F} = P = 2\left[-\frac{1}{12} + \frac{2}{45}\right] + \frac{2}{15} = -\frac{1}{6} + \frac{4}{45} + \frac{2}{15} = \frac{1}{18}$

$F = -18\,\text{cm}$. Focus is negative means system will behave as concave mirror.

2. ABCD

Sol. For maxima path difference = $n\,\lambda$

If $d$ = path difference between waves reaching point $O$ = $7\,\lambda$

*O* will be maxima.
For $d = \lambda$ only one maxima at *O* is possible, the screen being finite.

3. ABCD

Sol. $y = 2a \cos(\omega t + \phi) \cos(Kx)$ comparison with given equation gives

$K = \frac{2\pi}{\lambda} = 10\pi \Rightarrow \lambda = \frac{1}{5}m$

$\omega = 2\pi f = 50\pi \Rightarrow f = 25$ Hz

$\therefore$ $v = 5\ ms^{-1}$

at $x = 0.15$ m

$\cos(10\pi \times 0.15) = \cos(1.5\pi) = 0$ for all t

at $x = 0.3$

$\cos(10\pi \times 0.3) = \cos 3\pi = -1$ for all t

4. CD

Sol. If $\mu_1 = 0$, $\mu_2 \neq 0$

$t_1 = 0$, balancing torques about A

$N_1 \ell \sin\theta = mg\frac{\cos\theta}{2}$ ; $N_1 \tan\theta = \frac{mg}{2}$

If $\mu_1 \neq 0$, $\mu_2 \neq 0$

$T_2 = 0$, equilibrium cannot be attained.

If $\mu_1 \neq 0$, $\mu_2 \neq 0$

$N_1 = t_2 = \mu_2 N_2$ ; $N_2 + t_1 = mg$

$N_2 + \mu_1 N_1 = mg$ ; $N_2 + \mu_1(\mu_2 N_2) = mg$ ; $N_2 = \frac{mg}{1+\mu_1\mu_2}$

5. BC

Sol. $v_L = v_C = v_R$;

$\Rightarrow x_L = x_C = R$

when inductor is short circuited

$Z = \sqrt{R^2 + x_C^2} = \sqrt{2}\,R$

$\therefore I = \frac{30}{Z} = \frac{30}{\sqrt{2}R}$

$\therefore V_L = ix_L = \frac{30}{\sqrt{2}R} \times R = \frac{30}{\sqrt{2}}$

$\therefore$ (A) is incorrect and with similar calculations (B) will be correct.

Here $f_0$ is the resonance frequency as $v_L = v_C$

$\Rightarrow f_0 = \frac{1}{2\pi\sqrt{LC}}$

and $\omega_0 = 2\pi f_0 = \frac{1}{\sqrt{LC}}$

$\frac{x_L}{x_C} = \frac{\omega L}{1/\omega C} = \omega^2 LC$

Given $f = 2f_0$

$\Rightarrow \omega = 2\omega_0$

$\therefore \frac{X_L}{X_C} = 4$

$\therefore$ (C) is also correct.

6. ABD

Sol. For ammeter,

$i = \frac{i_{max}(R_s + R_A)}{(R_s)}$

$\Rightarrow$ $i = 0.1mA$ for $R_s = 50\Omega$ [as $R_A = 50\Omega$ and $i_{max} = 50\mu A$]

For voltmeter,

$V = i_{max}(R_A + R_V)$

$\Rightarrow$ $V \approx 10V$ for $R_v = 200k\Omega$

7. A

Sol. Time taken to move from mean to extreme (t) = $\sqrt{\frac{2A}{(F/m)}} = \sqrt{\frac{2Am}{F}}$

$\therefore$ Time period = $4t = 4\sqrt{\frac{2Am}{F}}$

8. A

Sol. Acceleration = constant $\Rightarrow$ velocity time curve should be straight line.

9. B

Sol. LC = $\frac{1}{10}$mm = 0.11 mm and zero error is $7 \times$ LC

10. C

Sol. Measured value $3.10 + 4 \times LC - 7 \times LC = 3.10 + 0.04 - 0.07 = 3.07$ cm.

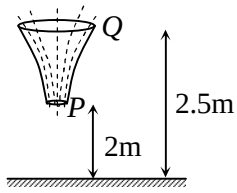
## SECTION – D

11. 01000.00

Sol. $P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) + \rho g(h_2 - h_1)$

$= \frac{1}{2} \times 10^3(1^2 - 3^2) + 10^3 \times 10(0.5)$

$= -4 \times 10^3 + 5 \times 10^3 = 1000$ J/m$^3$




12. 00000.20

Sol. $I_L$ and $I_C$ will be in opposite phases.

$I_{net} = I_L - I_C$

$= 0.6 - 0.4 = 0.2$ A

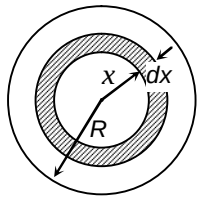
13. 00002.25

Sol. R to be maximum.

$$Y = \frac{3H}{2} = \frac{\text{Total height}}{2}.$$

14. 00002.50

Sol. $d\phi = (2\pi x dx)kxt^2$




$$\phi = \frac{2}{3}kt^2\pi x^3$$

$$\frac{d\phi}{dt} = \frac{4k\pi t x^3}{3}$$

$$E2\pi x = \frac{4\pi k t x^3}{3} \; ; \quad E = \frac{2}{3}ktx^2 \; ; \quad d\tau = \left(\frac{2}{3}ktx^2\right)\frac{Q}{\pi R^2}(2\pi x dx)x$$

$$\int d\tau = \frac{4}{3}\frac{ktQ}{R^2}\int_0^R x^4 dx \qquad \Rightarrow \quad \tau = \frac{4}{3}\frac{ktQ}{R^2}\frac{R^5}{5}$$

At $t$ = 15sec, $\tau$ = 2.50 Nm

15. 00000.30

Sol. $V_A = V_C$

$c = a + b$

16. 00000.10

Sol. $v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{4\times 1}{10\times 10^{-3}}} = 20 \quad m/s$

$$t = \frac{2\ell}{v} = 0.1\text{sec}$$

17. 00000.46

Sol. $W = \frac{V}{2}\rho_{Hg}g$ …(i)

$W = xV\,\rho_{Hg}\,g + (1 - x)\,V\,\rho_w\,g$ …(ii)

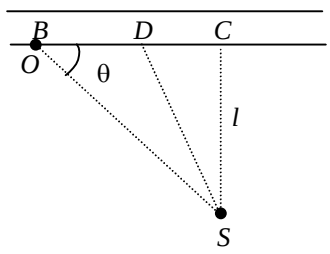
From (i) and (ii)

$$\frac{V}{2}\rho_{Hg}g = xV\rho_{Hg}g + (1-x)\,V\rho_w g$$

Solving we get x = 0.46

18. 00136.00

Sol. Let the bus be at O when it sends a signal that is detected by the detector as of frequency = 1500 Hz




$$\therefore \qquad f' = \left(\frac{v}{v - v_B\cos\theta}\right)\times 1000 = 1500$$

$$\therefore \qquad \cos\theta = \frac{\sqrt{3}}{2} \quad \Rightarrow \quad \theta = 30°$$

By the time signal reaches at S the bus reaches at D. Let this time be $t_0$

$$\therefore \qquad t_0 = \frac{OS}{v} = \frac{l\,\text{cosec}\,\theta}{v} \qquad \ldots (1)$$

Now man fires and the bullet reaches C in time $t_1$ (say). In the same time bus moves from D to C

$\therefore \quad t_1 = \frac{l}{u}$; where u = speed of bullet

Also, OD + DC = $l\cot\theta$

$v_B t_0 + v_B t_1 = l\cot\theta$

$v_B \left(\frac{l\operatorname{cosec}\theta}{v}\right) + v_B (l/u) = l\cot\theta$

$\frac{2}{3\sqrt{3}} \times 2 + \frac{2}{3\sqrt{3}} \frac{v}{u} = \sqrt{3}$

$\therefore \quad \frac{v}{u} = \frac{5}{2}$

$\therefore \quad u = \frac{2}{5} v = \frac{2}{5} \times 340 = 00136.00$ m/s

# Chemistry

## PART – II

### SECTION – A

19. ABD

Sol. In ABD the lone pair is less dispersed as compared to aniline.

20. ABD

Sol. Only (C) is asymmetric in nature, so active.

21. CD

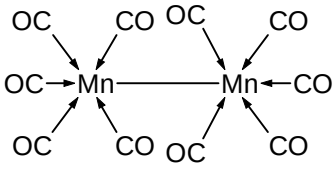
Sol. Ethers having – O – R group can be hydrolysed.

22. ACD

Sol. (A) z* increases from $Sc^{+}$ to $Sc^{3+}$ so I.E. increases.
(C) Down the group I.E. decreases
(D) Along the period I.E. increases

23. ACD

Sol. EAN of metal in A, C and D is equal to next inert gas.




24. AC

Sol. $NaH + H_2O \longrightarrow NaOH + H_2$

$$\underset{\text{(Molted)}}{NaH} \xrightarrow{\text{Electrolysis}} \underset{\text{(at cathode)}}{Na} + \underset{\text{(at anode)}}{\frac{1}{2}H_2}$$

25. C

Sol. In this case elimination reaction is dominating.

26. A

Sol. Position of E.W.G. ($NO_2$) decides the rearrangement step to form R – N = C = O.

27. A

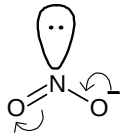
Sol. Extent of back bonding in B – F bond is decreased if – $NH_2$ or –OH groups are bonded with boron.

28. A

Sol. $O=\overset{+}{N}=\overset{-}{N}$ (Two types of bond)

$\overset{-}{N}=C=\overset{-}{N}$ (Both bonds are identical)

$\overset{-}{N}=\overset{+}{N}=\overset{-}{N}$ (Both bonds are identical)




(Both bonds are identical)

## SECTION – D

29. 00002.93

Sol. Fraction of edge unoccupied $= \dfrac{a-2R}{a}$

$a = 2\sqrt{2}\,R \qquad X = \dfrac{2\left(\sqrt{2}-1\right)}{2\sqrt{2}}$

$X = \dfrac{0.414}{1.414} = 0.293 \qquad \Rightarrow 10\,X = 2.93$

30. 00002.45

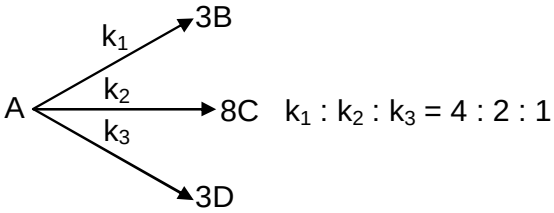
Sol. $\Delta T_f = iK_f m = 2 \times 1.86 \times 0.66$

31. 00002.60

Sol. $K_C = 10^{\frac{2(0.34-0.15)}{0.0591}} = 2.6 \times 10^6$

32. 00003.87

Sol.




$k_1 : k_2 : k_3 = 4 : 2 : 1$

mole of A remain after 45 days

$$= \frac{N_0}{2^n} = \frac{N_0}{2^{\frac{T}{t_{1/2}}}} = \frac{1}{2^{45/15}} = \frac{1}{2^3} = \frac{1}{8}$$

moles of [A] convert into product $= \dfrac{7}{8}$ mol

moles of $[B] = \dfrac{k_1}{k_1+k_2+k_3} \times \dfrac{7}{8} \times 3 = \dfrac{3}{2}$

moles of $[C] = \dfrac{k_2}{k_1+k_2+k_3} \times \dfrac{7}{8} \times \dfrac{8}{1} = 2$

and moles of $[D] = \dfrac{1}{7} \times \dfrac{7}{8} \times 3 = \dfrac{3}{8}$

33. 00001.63

Sol. $H_2CO_3 + NaOH \longrightarrow NaHCO_3 + H_2O$

$$pH_1 = \frac{1}{2}\left(pK_{a_1} + pK_{a_2}\right) = \frac{1}{2}(4.6+8) = 6.3$$

$N_1V_1 = N_2V_2$

$N_1 \times 10 = 0.1 \times 20$

$N_1 = 0.2$

$$NaHCO_3 + NaOH \longrightarrow Na_2CO_3 + H_2O$$

$$c = \frac{2m \text{ mole}}{50 \text{ mL}} = \frac{1}{25}M$$

$$\Rightarrow pH = \frac{[14 + 8 - 1.4]}{2} = 10.3$$

$$\frac{pH_2}{pH_1} = \frac{10.3}{6.3} = 1.63$$

34. 00000.80

Sol. Process AC = polytropic process

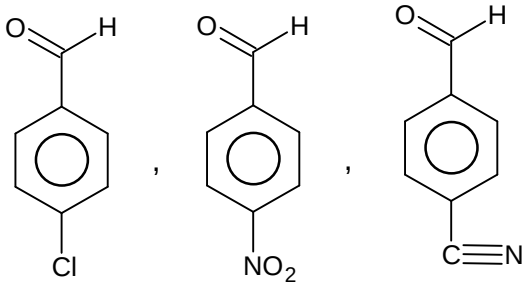
Molar Heat capacity $c_m = c_v + R/2 = 2R$

Process AB = Isobaric

$c_m = c_p = 5R/2$

$$\frac{q_{AC}}{q_{AB}} = \frac{\int_{T_A}^{T_C} nC_m \cdot dT}{\int_{T_A}^{T_B} n \cdot C_{p,m} \cdot dT} = \frac{2R}{\frac{5}{2}R} = 0.8$$

35. 00003.00

Sol. 


36. 00004.00

Sol. Graphite, $(BN)_x$, $B_3N_3H_6$, $C_6H_6$

# Mathematics

## PART – III

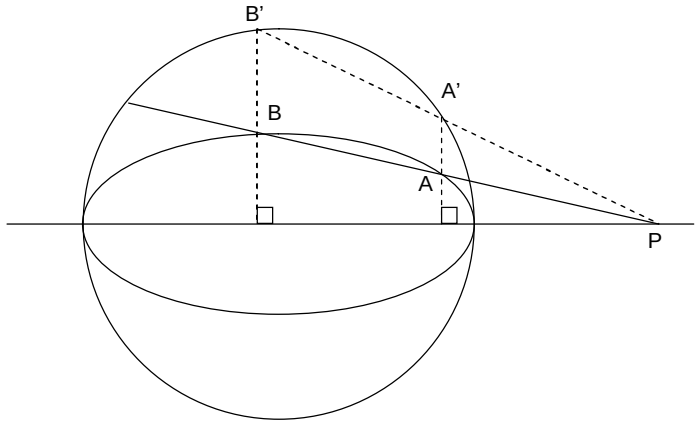
### SECTION – A

37. AB

Sol. Radical axis of $S_1 = 0$ and $S_2 = 0$ is the line $L_1 = 0$, the point of concurrency of family of lines $ax + by + c = 0$ if a,b,c are in A.P. is (1,-2)

38. AD

Sol. Take $A(\theta_1)$ and $B(\theta_2)$




39. BD

Sol. Equation of required plane is

$(x - y + 1) + \lambda(2y + z - 6) = 0$

$x + (2\lambda - 1)y + \lambda z + 1 - 6\lambda = 0$

Since it makes an angle of $30^o$ with x + y + z = 5

$$\Rightarrow \frac{\left|1 + (2\lambda - 1) + \lambda\right|}{\sqrt{3} \cdot \sqrt{1 + \lambda^2 + (2\lambda - 1)^2}} = \frac{\sqrt{3}}{2}$$

$$\Rightarrow |6\lambda| = 3\sqrt{5\lambda^2 - 4\lambda + 2}$$

$$\Rightarrow 4\lambda^2 = 5\lambda^2 - 4\lambda + 2 \Rightarrow \lambda = \left(2 \pm \sqrt{2}\right)$$

40. ABD

Sol. Volume $= \left|\left[2\vec{b} \times \vec{c} \quad 3\vec{c} \times \vec{a} \quad 4\vec{a} \times \vec{b}\right]\right| = 18$

$$\Rightarrow \quad 24\left[\vec{a} \quad \vec{b} \quad \vec{c}\right]^2 = 18$$

$$\Rightarrow \quad \left|\left[\vec{a} \quad \vec{b} \quad \vec{c}\right]\right| = \frac{\sqrt{3}}{2}$$

Now, $\left[\vec{a} \quad \vec{b} \quad \vec{c}\right] = \begin{vmatrix} (1+\sin\theta) & \cos\theta & \sin 2\theta \\ \sin\left(\theta + \frac{2\pi}{3}\right) & \cos\left(\theta + \frac{2\pi}{3}\right) & \sin\left(2\theta + \frac{4\pi}{3}\right) \\ \sin\left(\theta - \frac{2\pi}{3}\right) & \cos\left(\theta - \frac{2\pi}{3}\right) & \sin\left(2\theta - \frac{4\pi}{3}\right) \end{vmatrix}$

Applying $R_1 \to R_1 + R_2 + R_3$ and expanding

$$\left|\left[\vec{a} \quad \vec{b} \quad \vec{c}\right]\right| = \sqrt{3}\left|\cos 3\theta\right| = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \cos 3\theta = \pm\frac{1}{2} \Rightarrow 3\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3} \Rightarrow \theta = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{4\pi}{9}$$

41. CD

Sol. Clearly $p_1 = \frac{1}{2}$.

$$p_2 = p(HH) + p(T) = \frac{3}{4}$$

Now, the score n can be obtained in two distinct ways.
I. By throwing head when the score is n – 1
II. By throwing tail when the score is n – 2

$$\Rightarrow p_n = (p_{n-1})\frac{1}{2} + (p_{n-2})\frac{1}{2} = \frac{1}{2}(p_{n-1} + p_{n-2})$$

Also, from this $P_4 = \frac{11}{16}$.

42. B

Sol. $$\left(\frac{1}{(z-1)}(\cos\alpha + i\sin\alpha)\right)^{-1} = (z-1)(\cos\alpha - i\sin\alpha)$$

$\therefore$ $E = z_1 + \frac{1}{z_1}$ hence $I_m(E) = 0 \Rightarrow |z-1| = 1$

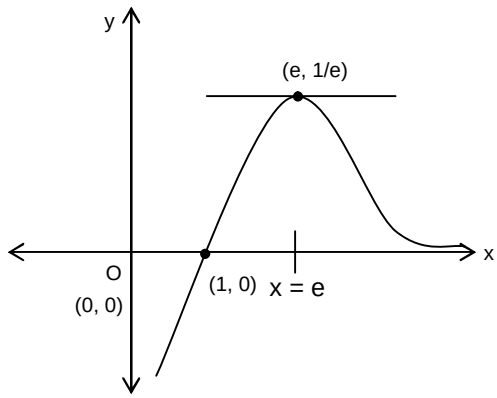
43. D

Sol. We have $b\log_e x - x = 0$

$$\Rightarrow \frac{\log_e x}{x} = \frac{1}{b} \qquad \ldots\ldots (1)$$

Let $y_1 = \frac{\log_e x}{x}$ and $y_2 = \frac{1}{b}$

Clearly, the equation $\frac{\log_e x}{x} = \frac{1}{b}$ has exactly one real root for positive real value of b = e.




44. D

Sol. Let $A(x_1, y_1)$ be the point of tangency.

Then $x_1 = b\log_e x_1 - x_1$ …….. (1)

Also, $1 = \frac{b}{x_1} - 1$ …….. (2)

$\therefore$ On solving (1) and (2), we get

$b = b\ln x_1 \Rightarrow b(\ln x_1 - 1) = 0$

$\therefore\ x_1 = e$ (As b > 0) (given)

So, $b = 2x_1 = 2e$

Also, point of contact is (e, e)

45. D

Sol. In the $\Delta ABC$, $AD \perp BC$ and $\angle BAC = \frac{2\pi}{3}$

$\Rightarrow |\underline{BAD} = |\underline{DAC} = \frac{\pi}{3}$

$\Rightarrow \Delta BAD$ and $\Delta DAC$ are congruent $\Delta$le's

$s \Rightarrow r_1 : r_2 = 1:1$

46. C

Sol. Let b > c

In $\Delta ADC$, $AD^2 + (a - x)^2 = b^2$ ....(1)

$\Delta ADC, AD^2 + x^2 = C^2$ ...(2)

Solving (1) and (2), $x = c\cos B$ and $AD = c\sin B$

From $\Delta ADC$

$$MD = S_1 - AC = \frac{a + b + c(\sin B - \cos B)}{2} - b$$

$$\Rightarrow MD = \frac{a - b + c(\sin B - \cos B)}{2}$$

From $\Delta ADB$

$$ND = S_2 - AB = \frac{c + c\cos B + c\sin B}{2} - c \Rightarrow ND = \frac{-c + c\cos B + c\sin B}{2}$$

$$\therefore\ MN = ME - ND = \frac{a - b + c - 2c\cos B}{2}$$

$$= R[\sin A - \sin B + \sin C - 2\cos B\sin C] = R[\sin(B + C) - \sin B + \sin C - 2\cos B\sin C]$$

$$= \left|4R\sin\left(\frac{B - C}{2}\right)\sin\frac{B}{2}\sin\frac{C}{2}\right|$$

## SECTION – D

47. 00008.00

Sol. $\text{Area} = \int_0^1 (6 - f(x))dx + \int_{-1}^0 (f(x) - (-2))dx = \frac{5}{4}$

48. 00008.00

Sol. $x^4 - 7x^2 - 4x + 20 = (x^2 - 4)^2 + (x - 2)^2$

$x^4 + 9x^2 + 16 = (x^2 + 4)^2 + x^2$

Take the curve $y = x^2$. Both square roots can be interpreted as distances.

49. 00001.00

Sol. $I=\int_0^{\pi} x\, f\left(\cos^2 x+\tan^4 x\right)dx$

$=\int_0^{\pi}(\pi-x) f\left(\cos^2(\pi-x)+\tan^4(\pi-x)\right)dx$

$=\int_0^{\pi}\pi f\left(\cos^2 x+\tan^4 x\right)dx-\int_0^{\pi} x\, f\left(\cos^2 x+\tan^4 x\right)dx$

$\Rightarrow \quad I=\frac{\pi}{2}\int_0^{\pi} f\left(\cos^2 x+\tan^4 x\right)dx$

$=\frac{\pi}{2}.2\int_0^{\pi/2} f\left(\cos^2 x+\tan^2 x\right)dx$

Hence $k=1$

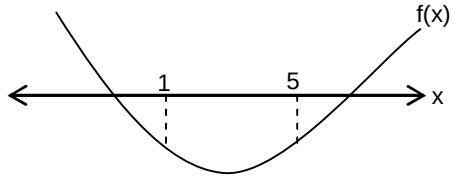
50. 00007.00

Sol. $\frac{(a+2)+(a+2)+b+c}{4}\geq\left((a+b)^2+bc\right)^{1/4}\geq(81)^{1/4}$

51. 00003.00

Sol. $XX^T=I$; Find PQ then $(PQ)^{10}$

52. 00005.00




Sol. We have $1+\sum_{r=1}^{10}\left(3^r.{}^{10}C_r+r.{}^{10}C_r\right)$

$=1+\sum_{r=1}^{10}3^r.{}^{10}C_r+10\sum_{r=1}^{10}{}^{9}C_{r-1}$

$=1+4^{10}-1+10.2^9$

$=4^{10}+5.2^{10}=2^{10}\left(4^5+5\right)=2^{10}\left(\alpha.4^5+\beta\right)$, so $\alpha=1$

and $\beta=5$

Now $f(1)<0$ and $f(5)<0$

So, $f(1)<0 \quad\Rightarrow\quad -k^2<0$

$\Rightarrow \quad k\neq 0$

and $f(5)<0 \quad\Rightarrow\quad 16-k^2<0$

$\Rightarrow \quad k^2-16>0$

$\Rightarrow \quad k\in(-\infty,4)\cup(4,\infty)$

Hence smallest positive integral value of k = 5

53. 00002.00

Sol. The given lines are $\frac{x-1}{1}=\frac{y-3}{-\lambda}=\frac{z-1}{\lambda}$ (=s)

and $\frac{x-0}{1/2}=\frac{y-1}{1}=\frac{z-2}{-1}(=t)$

are coplanar if $\begin{vmatrix} 0-1 & 1-3 & 2-1 \\ 1 & -\lambda & \lambda \\ \frac{1}{2} & 1 & -1 \end{vmatrix} = 0$

$$\Rightarrow \begin{vmatrix} -1 & -2 & 1 \\ 1 & -\lambda & \lambda \\ \frac{1}{2} & 1 & -1 \end{vmatrix} = 0$$

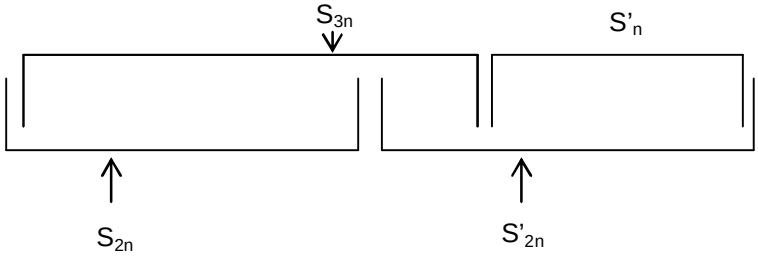
$$\Rightarrow 1(\lambda-\lambda)+2\left(-1-\frac{\lambda}{2}\right)+1\left(1+\frac{\lambda}{2}\right)=0$$

$$\Rightarrow \lambda = -2$$

54. 00005.00

Sol. Given $S_{3n} = S'_n = S_{4n} - S_{3n}$

$$\Rightarrow\ 2S_{3n} = S_{4n}$$




$$\Rightarrow 2\times\frac{3n}{2}\left[2a+(3n-1)d\right)=\frac{4n}{2}\left[2a+(4n-1)d\right]\Big]=$$

$$\Rightarrow 12a+(18n-6)d=8a+(16n-4)d$$

$$\Rightarrow 4a=(-2n+2)d \qquad \text{.....(i)}$$

$$\Rightarrow 2a=(1-n)d$$

Now, we have to find $\frac{S_{2n}}{S_{2n}'}$

$$\frac{S_{2n}}{S_{2n}'}=\frac{S_{2n}}{S_{4n}-S_{2n}}$$

$$=\frac{\frac{2n}{2}\left[2a+(2n-1)d\right]}{\frac{4n}{2}\left[2a+(4n-1)d\right]-\frac{2n}{2}\left[2a+(2n-1)d\right]}$$

$$=\frac{2\left[(1-n)d+(2n-1)d\right]}{4\left[(1-n)d+(4n-1)d\right]-2\left[(1-n)d+(2n-1)d\right]}=\frac{2^{nd}}{10^{nd}}=\frac{1}{5}$$